V9 FLOWCHART

CSE-CIC-IDS2018

Raw flows: 8,284,871 | Raw columns: 80 | 10 CSV day captures (AWS Cloud Traffic)

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Data Cleaning & Deduplication

• Remove identity / shortcut columns (Timestamp, IP identifiers, Dst Port)

• Remove ±Inf / NaN rows | Remove zero-variance features & duplicate records

8,284,871 → 6,320,003 clean flows (70 clean columns)

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Stratified Data Partitioning

80% Train Pool (5,056,002) / 20% Test Set (1,264,001)

Train Pool → 90% Core-Train (4,044,800) / 10% Dev Split (505,601)

• TEST: SEALED FROM TRAINING / FEATURE SELECTION

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Feature Preprocessing & Standardization

Protocol One-Hot Encoding → 69 dimensions

StandardScaler (Fitted strictly on Core-Train only)

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Mutual Information (MI) Selection

69 Encoded Dimensions → Top 40 Candidate Features

Mutual Information ranking (100K sample, MI fitted on Core-Train only)

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SMOTE Class Balancing

Core-Train only (Train-only) | sampling_strategy = 1.0 (1:1 Class Ratio)

4,044,800 → 7,191,590 balanced samples (40 features preserved)

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SHAP Adversarial-Stability Selection

40 MI Candidate Features → Clean SHAP + Adversarial SHAP (PGD, 20 steps, ε=0.05 on calibration proxy)

S(f) = 1 − |SHAP_clean(f) − SHAP_adv(f)| / (SHAP_clean(f) + SHAP_adv(f) + ε)

80% cumulative stability weight threshold → 27 robust features

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V9 Model Architecture (IDSNet)

Input: 27 features

Dense 128 → BatchNorm + ReLU + Dropout(0.30) → Dense 64 → BatchNorm + ReLU + Dropout(0.30)

Dense 32 → BatchNorm + ReLU + Dropout(0.30) → Output: 2 logits (Binary Classification)

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Phase 1 — Clean Pretraining

10 epochs on balanced Core-Train (7,191,590 samples) | Batch size: 2048 | Adam (lr=0.001) | Cross-Entropy

Output: Warm-start V9 model

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Phase 2 — Dynamic Online PGD Adversarial Training

40 epochs | Per-batch adversarial generation | PGD steps: 5 | α = 0.01 | ε = 0.05 | L∞ threat bound

Current model

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Current training batch → PGD generation → Adversarial batch

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Clean + Adversarial loss → Backpropagation → Optimizer update

↺ (Feedback Loop: Updates Current Model for Next Batch)

Joint Loss Objective: L_total = 0.5 L_clean + 0.5 L_adv

L_clean = CrossEntropy(clean predictions, labels) | L_adv = CrossEntropy(adversarial predictions, labels)

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Final V9 Defense Model

Input = 27 robust features | Output = Binary NIDS prediction

Multi-Threat Evaluation Protocol: 3 Seeds (42, 123, 2024) | ε = 0.05 | n = 2,000 test samples

A. Bounded White-Box

FGSM, PGD-100, BIM, MIM, APGD-CE, CW-L∞

B. Black-Box / Transfer

Square Attack, Transfer PGD (WiderNet)

C. Boundary Diagnostics

Boundary Search, Unconstrained DeepFool

AMD Radeon RX 7600 — 8 GB VRAM | PyTorch + ROCm